ifesserra-lab / horizon_etl

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paulossjunior docs: update backlog with PR 67 for Lattes ingestion — c74c086 · last month — 143 Commits

.agent	docs: update knowledge base, backlogs and pm artifacts	last month
.github	feat: Automatic Team Management with Fuzzy Matching (...	2 months ago
build/lib/src	feat: restore Lattes ingestion and fix conflicts	last month
cache	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
db	feat: Add articles export and improve serialization	last month
docs	docs: update backlog with PR 67 for Lattes ingestion	last month
horizon_etl.egg-info	feat: restore Lattes ingestion and fix conflicts	last month
scripts	mesage	last month
src	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
tests	feat: expanded advisorship ingestion and canonical export...	last month
.env.example	feat: research area mart, sigpesq idempotency and pipelin...	2 months ago
.flake8	feat: ensure agile standards compliance (docs, tests, env)	2 months ago
.gitignore	feat: implement parent project status calculation (RF-26)	2 months ago
Makefile	feat: add Serra-specific commands and configurable CAM...	2 months ago
README.md	docs: update README and Status Report with filtering feat...	2 months ago
app.py	feat: implement unified Lattes pipeline with complete ETL ...	last month
check_gustavo.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
check_schema.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
check_types.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
chromedriver	feat: restore Lattes ingestion and fix conflicts	last month
debug_db.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
docker-compose.yml	fix(infra): configure app.py for local prefect 3 server and u...	2 months ago
find_inconsistencies.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
gustavo_details.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
init_db.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
investigate_338.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
investigate_name.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month
lattes.config	feat: restore Lattes ingestion and fix conflicts	last month
pyproject.toml	feat: upgrade research-domain to v0.12.8 and fix downloa...	last month
requirements.txt	feat: refactor core to research-domain v0.12.7, add Co-Ad...	last month
test_db_connection.py	feat: research area mart, sigpesq idempotency and pipelin...	2 months ago
test_matcher.py	Refactor: Apply strategy pattern to Lattes ingestion (#67)	last month

README

Horizon ETL

Horizon ETL is a modern, scalable data pipeline built with **Python**, **Prefect**, and **Supabase**. It follows **Hexagonal Architecture** principles combined with a **Source/Transform/Sink** component model to ensure robustness, testability, and easy extensibility.

🏗️ Architecture

The project is structured to separate concerns strictly:

```
src/
├── core/                 # The "Hexagon" (Pure Business Logic)
│   ├── ports/            # Interfaces (Contracts for Sources/Sinks)
│   ├── domain/           # Data Schemas (Pydantic Models)
│   └── logic/            # Pure Transformations (Mappers/Cleaners)
│
├── adapters/             # The "Infrastructure" (Real World)
│   ├── sources/          # Implementations (FapesScraper, CnpqCrawler)
│   └── sinks/            # Implementations (SupabaseRepo, JsonSink, S3Bucket)
│
└── flows/                # The Orchestration (Prefect)
    └── sync_cnpq_groups.py # Orchestrates CNPq Sync
    └── export_canonical.py # Orchestrates Data Export
```

🚀 Key Features

- **Idempotency:** Proper handling of natural keys and Upserts to prevent data duplication.
- **Observability:** Integrated logging and Data Lineage via Prefect Artifacts.

About

ETL Pipeline for extracting, validating, and loading data from multiple sources (Web, API) into Supabase using Prefect and Hexagonal Architecture.

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Contributors 1

paulossjunior Paulo

Languages

Python 98.3% | Makefile 1.7%

- **Auditability:** Raw data storage for debugging and historical replay.
- **Extensibility:** Plug-and-play architecture using Ports & Adapters.
- **CNPq Synchronization:**
  - Automated extraction of Research Groups from Lattes/CNPq.
  - **Research Lines:** Maps CNPq 'linhas_de_pesquisa' to 'Knowledge Areas'.
  - **Self-Healing:** Robust researcher recovery and association.
- **Canonical Data Export:** Generates standardized JSON dumps for Organizations, Campuses, Knowledge Areas, Researchers, and Research Groups.

🛠️ Stack

- **Orchestrator:** Prefect 3
- **Database:** Supabase (PostgreSQL)
- **Language:** Python 3.10+
- **Validation:** Pydantic

📦 Getting Started

1. Installation

```
git clone https://github.com/ifesserra-lab/horizon_etl.git
cd horizon_etl
pip install -e .
```

2. Running Flows

Sync CNPq Research Groups:

```
# Sync specific campus (e.g., Serra)
python app.py cnpq_sync Serra
```

Export Canonical Data:

```
# Exports to data/exports/
python app.py export_canonical

# Export with Campus Filter
python app.py export_canonical data/exports "Serra"
```

Generate Knowledge Area Mart:

```
# Generate Mart JSON
python app.py ka_mart

# Generate Mart filtered by Campus
python app.py ka_mart data/exports/mart.json "Serra"
```

📜 Version History

- **v0.5.0**: CNPq Sync Enhanced (Research Lines & Knowledge Areas), Canonical Exports, Fix for Researcher Persistence.
- **v0.4.0**: Base CNPq Synchronization.
- **v0.3.0**: SigPesq Enhancements & Granular Strategy Pattern.
- **v0.2.0**: Research Group Ingestion & Local Infrastructure.

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